Lecture 39: Deep Learning (Part-4) — Practice Problems Suite (Hinglish)

Yeh problem suite Computer Vision, Convolutional Neural Networks, mathematical convolutions, spatial geometry (padding aur stride), pooling layers, aur end-to-end CNN architectures par based hai.

Problem 1: Deep Learning Taxonomy aur Inductive Biases

Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNN) aur Recurrent Neural Networks (RNN) ka comparison karein. Har model apne specific data format (Tabular, Images, Sequential Text/Audio) ke liye hi kyun best suited hota hai?

Problem 2: Digital Image Matrix Representation

Ek camera 1920×1080 resolution ki color photo capture karta hai:

1. Is tensor ki shape channels-last (H, W, C) aur channels-first (C, H, W) format mein likhein.
2. Is single photo mein total kitne numeric pixel values hain?
3. Pure Yellow, Pure Cyan aur 50% Gray color ke RGB pixel values kya honge?

Problem 3: Pixel Normalization Ki Zaroorat

Raw pixels $[0, 255]$ ko neural network mein pass karne se pehle 255.0 se divide karke $[0.0, 1.0]$ floating range mein convert karna kyun zaroori hai? Agar bina normalize kiye raw integers pass karein toh kya hoga?

Problem 4: Computer Vision Ke 4 Mukhya Tasks

Computer Vision ke 4 primary tasks ka comparison karein:

1. Image Classification
2. Classification with Localization
3. Object Detection
4. Semantic Segmentation Har task ka exact mathematical output (labels, bounding boxes, pixel masks) kya hota hai?

Problem 5: Autonomous Vehicles Mein Multi-Task Vision

Self-driving cars (jaise Tesla ya Waymo) live camera streams par ek sath multiple computer vision tasks kaise run karti hain taaki sadak, gaadiyan, pedestrians, stop signs aur gaddhe (potholes) sab ek sath detect ho sakein?

Problem 6: FNN Mein Parameter Explosion Ka Ganit

Maan lijiye hum ek full-HD RGB image ($1920 \times 1080 \times 3$) ko standard ANN mein pass karte hain. Agar pehli hidden layer mein 256 neurons hon:

1. Is single layer ke total weights aur biases calculate karein.
2. Yeh model computer memory aur overfitting ke hisab se kyun fail ho jayega?

Problem 7: Spatial Locality aur Translation Invariance

Spatial Locality aur **Translation Invariance** kya hoti hain? 2D matrix ko khinch kar 1D vector bana dene se padosi pixels ka aapas ka relation kaise khatam ho jata hai?

Problem 8: CNN Ka Do-Stage Blueprint

CNN ke do foundational stages ka diagram banakar explain karein:

- Stage 1: Feature Extraction Engine
- Stage 2: Classification Head Stage 1 aur Stage 2 ke beech mein data kis form mein transfer hota hai?

Problem 9: Hierarchical Visual Feature Learning

CNN biological human visual cortex ki tarah hierarchical learning kaise karta hai? Shuruati layers (edges/lines), beech ki layers (textures/shapes) aur aakhri layers (object parts) ka flow samjhayein.

Problem 10: 2D Convolution Ka Manual Calculation

Diye gaye 4×4 input X aur 2×2 kernel K (bias $b = 1$) ke liye:

$$X = \begin{bmatrix} 2 & 0 & 1 & 3 \\ 1 & 2 & 0 & 2 \\ 0 & 1 & 3 & 1 \\ 2 & 1 & 0 & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad b = 1$$

Stride $S = 1$ aur Valid Padding $P = 0$ maankar:

1. Output feature map ka size kya hoga?

2. Top-left cell $(0, 0)$ aur top-right cell $(0, 2)$ ki exact numeric output values calculate karein.

Problem 11: Handcrafted Sobel Filters vs Learnable CNN Kernels

Purane zamane ke fixed Sobel filters aur modern deep learning CNN kernels mein kya farq hai? CNN ke filter ke numbers training ke dauran kaise decide hote hain?

Problem 12: Output Dimension Ka Universal Formula

Input Width W, Kernel Size K, Padding P, aur Stride S ke sath output size O nikalne ka formula likhein.

1. $W = 32, K = 5, P = 2, S = 1$ ka output size nikalein.
2. $W = 224, K = 7, P = 3, S = 2$ ka output size nikalein.
3. Prove karein ki Same Padding ($P = (K - 1)/2$) lagane par output size input size ke barabar kyun rehta hai.

Problem 13: Pooling Layer Ke Niyam

Deep CNNs mein Pooling Layer kyun lazmi hoti hai?

1. Neeche diye gaye 4×4 matrix par 2×2 Max Pooling (stride 2) lagakar output nikalein:

$$\begin{bmatrix} 4 & 9 & 1 & 3 \\ 8 & 2 & 7 & 5 \\ 3 & 1 & 14 & 10 \\ 6 & 0 & 2 & 12 \end{bmatrix}$$

2. Prove karein ki pooling layer mein learnable parameters ZERO kyun hote hain ($\text{params} = 0$).
3. Max Pooling translation invariance kaise provide karta hai?

Problem 14: Flattening, Dense Reasoning aur Dropout

1. Agar aakhri pooling layer $(128, 7, 7)$ tensor deti hai, toh `Flatten()` karne par 1D vector ki length kitni hogi?
2. Fully connected layer ko "global reasoning" layer kyun kaha jata hai?
3. Dropout ($p = 0.5$) dense layer mein overfitting kaise rokta hai?

Problem 15: Poore CNN Network Ka Parameter Calculation

Ek complete CNN architecture ka layer-by-layer audit karein:

- Input: $(3, 32, 32)$ (CIFAR-10 RGB image)
- Layer 1: `Conv2d(3, 32, kernel_size=3, padding=1, stride=1)` + `ReLU`
- Layer 2: `MaxPool2d(kernel_size=2, stride=2)`
- Layer 3: `Conv2d(32, 64, kernel_size=3, padding=1, stride=1)` + `ReLU`
- Layer 4: `MaxPool2d(kernel_size=2, stride=2)`
- Layer 5: `Flatten()`
- Layer 6: `Linear(in_features=?, out_features=128)` + `ReLU`
- Layer 7: `Linear(128, 10)` Calculate karein:

1. Har layer ke baad output tensor shape.
2. Har layer ke learnable parameters.
3. Poore network ke total trainable parameters.